

**INTERNSHIP PROJECT REPORT**  
**ON**

**“Design and Development of a Secure User ID Creation and Onboarding System for Banks, Financial Intermediaries, and Other Entities on the NCRP Portal for Citizens’ Financial Cyber Fraud Reporting and Management”**

**Submitted To**

**Cyber Fraud Mitigation Center,  
Indian Cyber Crime Coordination Center,  
Ministry Of Home Affairs,  
Government Of India**

**For fulfilment of the internship  
01<sup>st</sup> Dec-2025 to 02<sup>nd</sup> April 2026**

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I would like to thank my supervisors and mentors at MHA I4C, particularly the Director, CFMC, **Lt. Col. Badal Kaushik, Dy. Director Akhilesh Gaur, Mr. Lavish Yadav, Mrs. Sneha Singh, Mrs. Priyanka Kumari** for his tremendous guidance, support, and encouragement throughout my internship. This work significantly strengthened my understanding of cybersecurity principles, secure system architecture, and access governance mechanisms. It provided valuable technical exposure to secure authentication, role-based access control, database security, and structured workflows for financial cyber fraud management, while also enhancing my analytical and technical competencies in designing secure and scalable systems.

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With Sincere Regards,

**Prashant Gond**

## **ABSTRACT**

The rapid growth of digital financial services has led to a significant increase in financial cyber frauds, necessitating a secure, efficient, and centralized reporting and management mechanism. The National Cyber Crime Reporting Portal (NCRP) plays a critical role in enabling citizens to report financial cyber fraud incidents. However, effective coordination with banks, financial intermediaries, and other authorized entities requires a robust and controlled onboarding and user identity management system.

This project focuses on the **design and development of a User ID creation and onboarding system** for banks, financial intermediaries, and other entities within the NCRP ecosystem. The system provides a structured workflow for entity registration, document submission, verification, and approval through an administrative interface. Role-based access control (RBAC) ensures that only authorized personnel can access sensitive fraud-related data, thereby enhancing security and accountability.

The solution integrates secure authentication mechanisms, validation checks, audit logging, and automated approval/rejection workflows to streamline onboarding while maintaining regulatory compliance. Additionally, the system supports scalability and interoperability to accommodate a growing number of participating entities. By strengthening institutional onboarding and access governance, the proposed system improves response efficiency, inter-agency collaboration, and overall effectiveness of financial cyber fraud reporting and management within the NCRP framework.

**Keywords:** *Financial Cyber Fraud, National Cyber Crime Reporting Portal (NCRP), User ID Management, Entity Onboarding System, Banks and Financial Intermediaries, Role-Based Access Control (RBAC), Secure Authentication, Fraud Reporting and Management, Access Governance, Cybersecurity, Digital Identity Management, Administrative Workflow.*

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## **LIST OF ABBREVIATIONS**

### **Abbreviation**

### **Description**

**NCRP**

National Cyber Crime Reporting Portal

**I4C**

Indian Cyber Crime Coordination Centre

**MHA**

Ministry of Home Affairs

**FI**

Financial Intermediary

**RBAC**

Role-Based Access Control

**UID**

User Identification

**OTP**

One-Time Password

**API**

Application Programming Interface

**GUI**

Graphical User Interface

**DB**

Database

**CFMC**

Cyber Fraud Mitigation Centre

**CFCFRMS**

Citizens Financial Cyber Fraud Reporting & Management System

## **Chapter 1:**

### **Task :1 Understanding the National Cybercrime Reporting Portal (NCRP)**

#### **1.1 Introduction**

The National Cybercrime Reporting Portal (NCRP) initiative by the Government of India enables citizens to report cybercrimes online. It plays a crucial role in helping victims of cyber fraud, financial scams, online harassment, and other digital offences.

#### **1.2. Purpose of NCRP**

The primary objectives of the NCRP are:

- Providing a centralized platform for reporting cybercrimes.
- Assisting law enforcement agencies in tracking and investigating cybercriminals.
- Offering a structured complaint registration process for victims.

#### **1.3. Categories of Cybercrimes Handled**

The NCRP allows users to report various cybercrimes, including:

1. Financial Frauds – Online banking frauds, UPI scams, debit/credit card fraud, phishing.
2. Online Harassment and Abuse – Cyberstalking, blackmailing, image/video morphing, defamation.
3. Women and Child-Related Cybercrimes – Child pornography, sexual exploitation, cyberbullying.
4. Other Cybercrimes – Hacking, identity theft, cyber terrorism, fake social media profiles.

#### **1.4. How to File a Complaint**

Users can report a cybercrime through the NCRP portal by following these steps:

1. Visit the official NCRP website.
2. Select the category of cybercrime.
3. Provide details of the incident, including evidence such as screenshots, transaction records, or messages.
4. Submit the complaint and note down the unique complaint reference ID.
5. Track complaint status using the reference ID.

#### **1.5. Role of Law Enforcement Agencies**

- NCRP complaints are assigned to relevant state or central cybercrime units.
- Investigators analyze evidence and liaise with financial institutions or technology providers for further details.
- Law enforcement uses forensic techniques to trace and apprehend cybercriminals.



## 1.6. Financial Fraud Handling

- Victims must provide transaction details, screenshots, or bank statements.
- NCRP works with banks and financial institutions to freeze fraudulent transactions.
- Cases are prioritized based on severity and monetary loss.

## 1.7. Privacy and Security Measures

- NCRP ensures data confidentiality using encryption and OTP authentication.
- Complainants can report sensitive cases anonymously.

## 1.8. Support and Helpline Services

- 24/7 national cybercrime helpline for emergency assistance.
- Live chat support for guidance on filing complaints.
- Users can check their complaint progress via the portal.

## 1.9. Conclusion

The NCRP is a significant step towards strengthening cybersecurity and providing victims with a reliable platform for justice. It bridges the gap between victims and law enforcement, ensuring prompt action against cyber offenders.

- Below are some images regarding the NCRP Portal

Fig 1.9.1 shows the interface of the NCRP Portal

The screenshot displays the National Cyber Crime Reporting Portal. At the top, there is a header with the Government of India and Ministry of Home Affairs logos, the portal's name in Hindi and English, and a language selection dropdown. The main content area is titled "Filing a Complaint on National Cyber Crime Reporting Portal" and contains a disclaimer and terms of use. Below this, there is a "Useful Links" section with icons for various government services and organizations.

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National Cyber Crime Reporting Portal

Language

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अमृत महोत्सव

**Filing a Complaint on National Cyber Crime Reporting Portal**

Prior to filing a complaint with this portal, we would request you to read the below information regarding terms and conditions. Should you have queries prior to filing your complaint, view [Frequently Asked Questions FAQ](#)

The information I've provided on this form is correct to the best of my knowledge. I acknowledge that providing false information could make me liable to penal actions under Indian Laws.

I understand that action on the complaints reported on this portal shall be taken by concerned authorities as per Indian Laws.

The complaint information you submit to this site is encrypted via secure socket layer (SSL) encryption. Please see the [Privacy Policy](#) for further information.

We thank you for your cooperation.

**Useful Links**

india.gov.in

CTAT

CyTrain

Information Security  
Education & Awareness

कर्मयोगी भारत

NIELIT

Fig 1.9.2 Citizens login and file their complaints

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Language



HOME

LOGIN

CHECK LIST FOR COMPLAINANT

Please keep this information ready before filing your complaint:

Mandatory Information

- Incident Date and Time.
- Incident details (minimum 200 characters) without any special characters (#\$%^&\*~!@).
- Soft copy of any national Id (Voter Id, Driving license, Passport, PAN Card, Aadhar Card) of complainant in .jpeg, .jpg, .png format (file size should not more than 5 MB).
- In case of financial fraud, please keep following information ready:
  - Name of the Bank/ Wallet/Merchant
  - 12-digit Transaction id/UTR No.
  - Date of transaction
  - Fraud amount
  - Soft copy of all the relevant evidences related to the cyber crime (not more than 10 MB each)

Optional/Desirable Information:

- Suspected website URLs/ Social Media handles (wherever applicable)
- Suspect Details (if available)
  - Mobile No
  - Email id
  - Bank Account No
  - Address
  - Soft copy of photograph of suspect in .jpeg, .jpg, .png format (not more than 5 MB)
  - Any other document through which suspect can be identified.

CITIZEN LOGIN

[Click Here for Existing User](#)

SELECT STATE: \*

---Select---

LOGIN ID: \*

(Preferably your Email Id.)

Preferably your Email Id.

MOBILE NO: \*

+91

Mobile No.

Get OTP

OTP: \*

Your OTP Number

Enter Captcha

SUYN8p



Clear

Submit

[Forgot Login Id](#)

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Draft Complaint

Report Cyber Crime

Check Status

Complaint Withdraw

Update Mobile Number

User Profile

USER PROFILE DETAILS

Login Id :

9064421877prashant@gmail.com

Title\* :

Mr

Mrs

Dr

Shri

Smt

Prof

Miss

Name : \*

Prashant Gond

Mobile : \*

8001354567

DOB : \*

20 / 08 / 2004

Gender : \*

Male

Email Id : \*

9064421877prashant@gmail.com

Father/Mother/Spouse : \*

Select

Father/Mother/Spouse name

PRESENT ADDRESS

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**Fig 1.9.3 Here, the citizen fill in their details and tell what type of fraud happens**

The screenshot shows the 'PRESENT ADDRESS' form on the National Cyber Crime Reporting Portal. The form is divided into two columns. The left column contains fields for House No., Street Name, Colony, Vill/Town/City, and Tehsil. The right column contains fields for Country, State, District, Police Station, and Pincode. All fields are filled with sample data. Below the form is an 'Update' button and a note about allowed characters.

|                |                          |                |                                     |
|----------------|--------------------------|----------------|-------------------------------------|
| House No.      | Opp. Umang Marriage Hall | Country        | INDIA                               |
| Street Name    | Jhanakpura road          | State *        | WEST BENGAL                         |
| Colony         | Jhanakpura               | District*      | ASANSOL DURGAPUR POLICE COMMISSIONE |
| Vill/Town/City | Asansol                  | Police Station | KULTI                               |
| Tehsil         | barabani                 | Pincode *      | 713343                              |

(AlphaNumeric and Symbols like @ , . ( ) / \_ - : are allowed, Do not use any special characters.)

**Fig:1.9.4 Here, share the details, and citizens can track their complaints also**

The screenshot shows the 'Incident Details' form on the National Cyber Crime Reporting Portal. The form has a header with navigation tabs: Incident Details, Suspect Details, Complainant Details, and Preview & Submit. The main form area is titled 'Complaint / Incident Details'. It contains fields for Category of complaint, Sub-Category of complaint, Approximate date & time of Incident/receiving/viewing of content, Is there any delay in reporting?, Where did the incident occur?, and Please provide any additional information about the incident. A dropdown menu for Sub-Category of complaint is open, showing various fraud types. A note at the bottom right specifies character requirements.

**Category of complaint\***: Online Financial Fraud

**Sub-Category of complaint \***: ---Select---

**Approximate date & time of Incident/ receiving/viewing of content \***: [Date/Time]

**Is there any delay in reporting?**: [Yes/No]

**Where did the incident occur? \***: [Location]

**Please provide any additional information about the incident \***: [Text Area]

(Insert at least 200 Characters. Special Characters like ~! #^`\$|{}<>\* are not allowed)

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Language



Home

Register a Complaint +

Track your Complaint

Report & Check Suspect +

Cyber Volunteers +

Learning Corner +

Citizen Survey

Contact Us

Home > Track your Complaint

Track your Complaint Status

Acknowledgement No. \*

Enter Acknowledgement No.

Get OTP

Enter OTP here...

b3FZUG

Enter Captcha

Submit

Useful Links

















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## **Chapter 2:**

### **Design and Development of a Secure User ID Creation and Onboarding System for Banks, Financial Intermediaries, and Other Entities to Enable Controlled Access to the NCRP Portal**

#### **2.1 Introduction**

The National Cyber Crime Reporting Portal (NCRP) is a critical national platform for reporting and managing cybercrime incidents, particularly financial cyber fraud. To ensure effective coordination between citizens, law enforcement agencies, banks, financial intermediaries (FIs), and other authorized entities, it is essential to provide a secure and well-governed access mechanism.

This chapter focuses on the design and development of a secure User ID creation and onboarding system for banks, financial intermediaries, and other entities. The system enables controlled access to the NCRP portal while ensuring data confidentiality, integrity, and accountability. A structured onboarding process ensures that only verified and authorized entities can access sensitive financial cyber fraud information.

#### **2.2 Need for a Secure Onboarding System**

Financial cyber fraud cases involve sensitive personal, financial, and transactional data. Unrestricted or weak access controls may lead to data leakage, misuse of information, or unauthorized actions. Therefore, a secure onboarding and User ID management system is required to:

- Authenticate legitimate banks, financial intermediaries, and authorized entities
- Prevent unauthorized access to sensitive NCRP data
- Maintain accountability through user-level tracking and audit logs
- Enable faster response and coordination during financial fraud incidents

- Ensure compliance with government cybersecurity and data protection guidelines

### 2.3 Stakeholders Involved in the Onboarding Process

The onboarding system is designed to support multiple stakeholders involved in financial cyber fraud reporting and management:

- **Banks:** To verify transactions, freeze fraudulent accounts, and assist victims
- **Financial Intermediaries (FIs):** Payment gateways, wallets, NBFCs, and fintech platforms
- **Other Entities:** Telecom service providers, technology partners, and regulatory bodies
- **NCRP Administrators:** Responsible for verification, approval, and access governance

Each stakeholder is provided access strictly based on their role and responsibilities.

### 2.4 User ID Creation Workflow

The User ID creation process follows a structured and secure workflow to ensure authenticity and accountability.

#### Steps involved in User ID creation:

1. Entity registration request submission on the NCRP portal
2. Upload of required documents such as authorization letters, identity proof, and official credentials
3. Validation of submitted details at the system level
4. Administrative review and verification by NCRP authorities
5. Approval or rejection of the onboarding request
6. Secure generation of User ID credentials for approved entities

This workflow minimizes human error and ensures only verified entities are onboarded.

### 2.5 Entity Onboarding and Document Verification

Document verification is a crucial part of the onboarding system. Entities are required to submit officially approved documents, which may include:

- Organization registration certificates
- Authorization letters issued by competent authorities
- Official email domain verification
- Identity proof of authorized personnel

The NCRP administrative panel enables officers to review documents, request clarifications, and approve or reject onboarding requests. This ensures transparency and traceability in the onboarding process.

## **2.6 Role-Based Access Control (RBAC)**

Role-Based Access Control (RBAC) is implemented to restrict access based on the user's role. Each User ID is assigned specific permissions depending on the entity type and responsibility.

Examples of roles include:

- Bank User
- Financial Intermediary User
- Administrative User
- Supervisory User

RBAC ensures that users can only access information relevant to their role, reducing the risk of unauthorized data exposure.

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## **2.7 Authentication and Security Mechanisms**

To safeguard access to the NCRP portal, multiple security mechanisms are integrated into the User ID system:

- Secure login credentials
- One-Time Password (OTP) based authentication
- Encrypted communication using secure protocols
- Password complexity and expiry policies

These measures help protect against unauthorized access, credential theft, and cyber attacks.

## **2.8 Administrative Approval and Monitoring**

An administrative approval mechanism is built into the system to ensure oversight and governance. NCRP administrators can:



- View onboarding requests
- Verify submitted documents
- Approve or reject User ID requests
- Monitor user activities through logs

Audit logs record login attempts, data access, and actions performed by users, ensuring accountability and compliance.

## 2.9 System Architecture Overview

The onboarding system follows a layered architecture:

- **Presentation Layer:** Web-based GUI for entity registration and administration
- **Application Layer:** Handles business logic, validation, and workflows
- **Database Layer:** Stores user details, roles, logs, and onboarding records

This architecture ensures scalability, maintainability, and security.

## 2.10 Conclusion

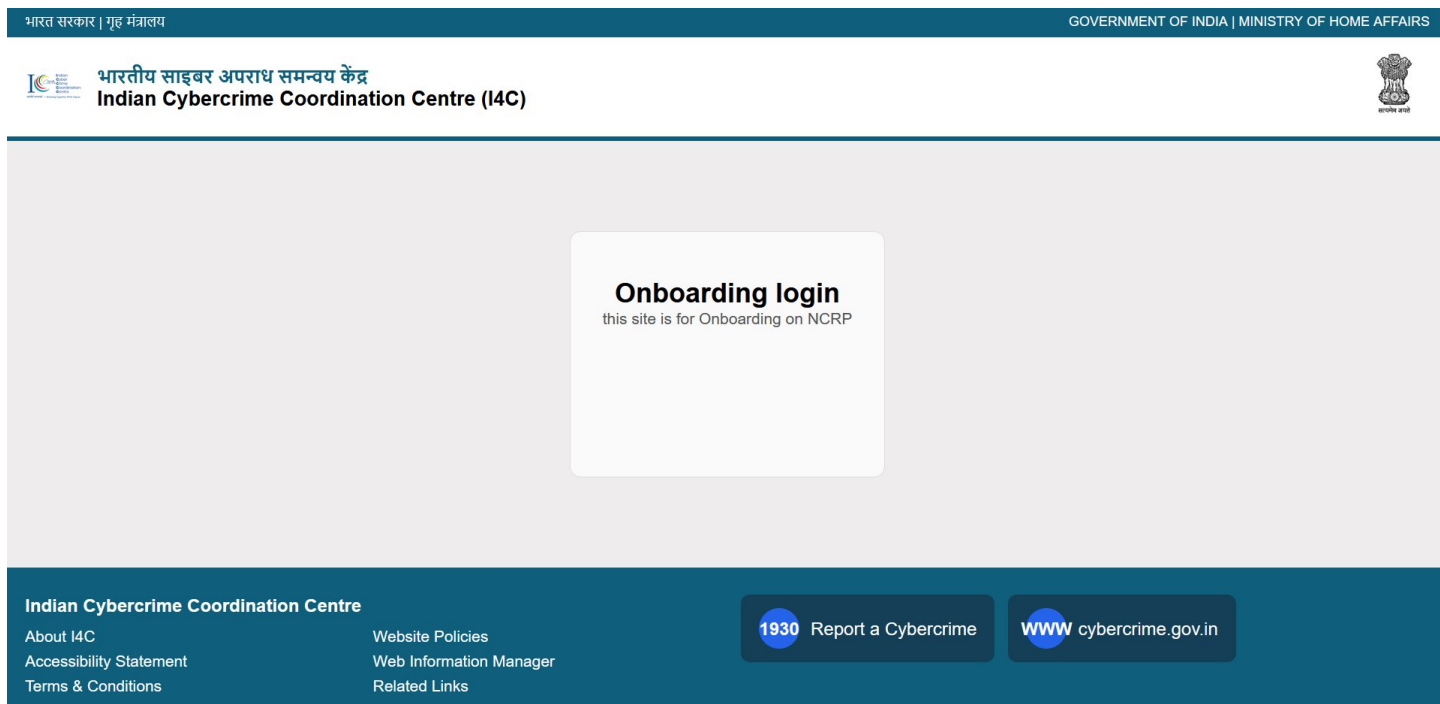
The secure User ID creation and onboarding system plays a vital role in strengthening the NCRP portal's operational efficiency. By ensuring controlled access, robust authentication, and role-based authorization, the system enhances trust, coordination, and response effectiveness in financial cyber fraud reporting and management. The proposed design supports scalability and future enhancements while aligning with national cybersecurity objectives.



## 1. Onboarding Login Page

This page represents the onboarding login interface of the NCRP portal developed under I4C.

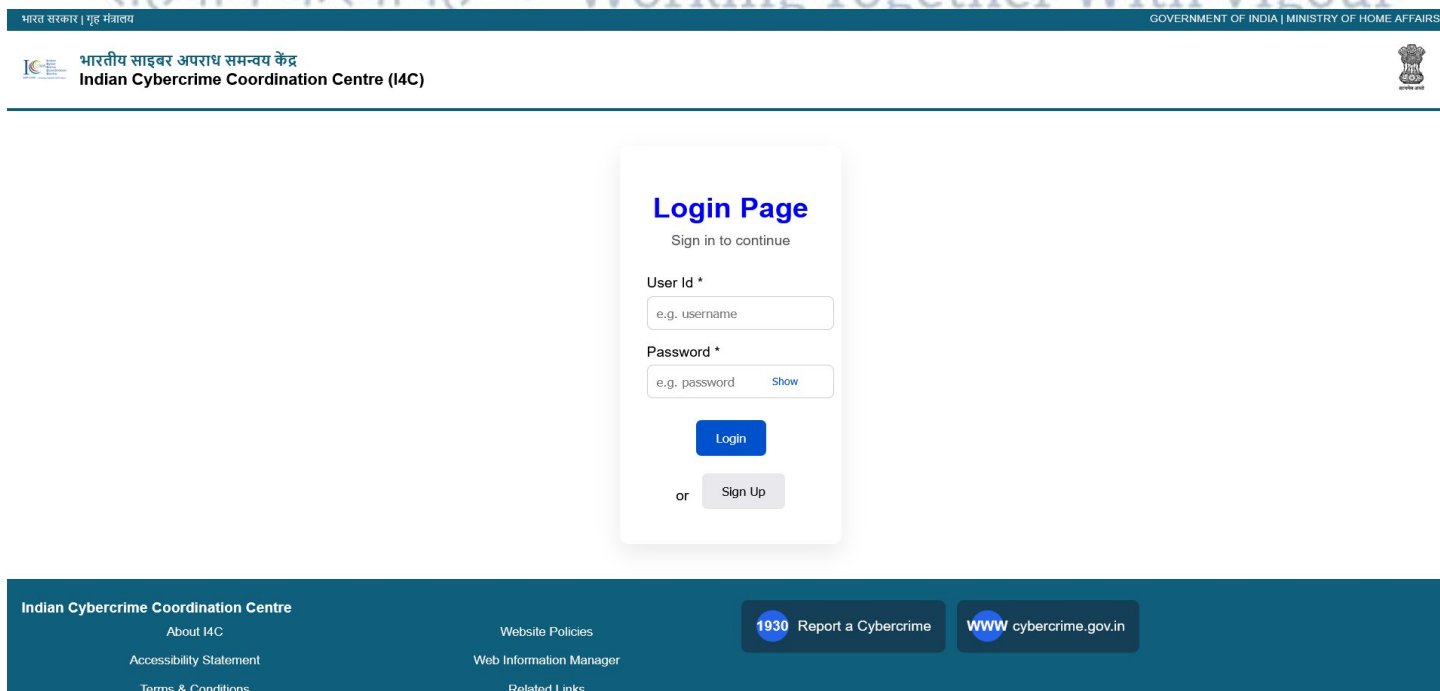
It serves as the secure entry point for banks, financial intermediaries, and other entities to access the onboarding system.



## 2. User Login Page

This page allows registered users to log in using their User ID and password.

Secure authentication ensures that only authorized users can access NCRP onboarding services.



### 3. Authorized Bank Official Details Page

This page captures personal and official details of the authorized bank official.

The information is used to create a unique User ID and ensure accountability of the registered user.

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Indian Cybercrime Coordination Centre (I4C)



### Authorized Bank Official Details

Full Name \*

Prashant Shah Gond

Official Email ID \*

prashant.gond100@gmail.com

User Email ID \*

9064421877prashant@gmail.com

Designation \*

Intern

Official Contact Number \*

9876543210

Create User ID \*

prashant.gond100@gmail.com

Create Password \*

\*\*\*\*\*

Submit & Continue

Reset



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### 4. Registration of Banks / FIs / Other Entity Page

This page facilitates the registration of banks, financial intermediaries and other entities on the NCRP portal.

Users are required to provide organizational and regulatory details as per RBI guidelines.

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### ONBOARDING OPTIONS



**Bank Registration for Onboarding**  
Register banks for onboarding on NCRP



**FI Registration for Onboarding**  
Register financial institutions for onboarding



**Other Entities Registration**  
Register other organizations for onboarding

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## 5. Registration of Banks Page

This page facilitates the registration of banks on the NCRP portal.

Users are required to provide organizational and regulatory details as per RBI guidelines.



### Registration of Banks

Please fill all required details to get onboarded

Select Bank Category:

-- Select Category --

Select Bank / FI \* (as per RBI list)

Select bank category first

#### Regulatory Documents

Date of Incorporation

dd / mm / yyyy

#### Regulatory Documents

RBI Registration Certificate (PDF/JPG)

Browse... No file selected.

GST Certificate- 3 pages

Browse... No file selected.

FIU Registration ID

#### Regulatory Documents

RBI Registration Certificate (PDF/JPG)

Browse... No file selected.

GST Certificate- 3 pages

Browse... No file selected.

FIU Registration ID

REID1234

FIU\_Registration Screenshot

FIU sample

[Download FIU sample](#)

Browse... No file selected.

#### Financial Information

Balance Sheet – Year 1

Browse... No file selected.

Balance Sheet – Year 2

Browse... No file selected.

Nodal officer sheet :

Nodal\_sample

[download nodal sample](#)

Browse... No file selected.

Balance Sheet – Year 1

No file selected.

Balance Sheet – Year 2

No file selected.

Please select a file.

Nodal officer sheet :

Nodal\_sample

[download nodal sample](#)

No file selected.

Upload Terms and Conditions :

TnC sample

[Download TnC sample](#)

No file selected.



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## 6. Financial Intermediaries Registration

This page is used for registering financial intermediaries involved in payment systems. It captures regulatory, financial, and operational details required for NCRP access.

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## Registration of FIs - Setting up and Operating Payment System

Please fill all required details to get onboarded

Select FI Category:

Financial Market Infrastructure

Select FI \*

The Clearing Corporation of India Ltd

Date of Incorporation

03 / 01 / 2018

RBI Registration Certificate

No file selected.

GST Certificate



No file selected.

FIU Registration ID

FIU Registration Screenshot  
 No file selected.

**Financial Information**

Balance Sheet – Year 1  
 No file selected.

Balance Sheet – Year 2  
 No file selected.

Balance Sheet – Year 3  
 No file selected.

Nodal Officer Sheet  
 No file selected.

Upload Terms & Conditions  
 No file selected.



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## 7. Registration for Other Entities Page

This page enables onboarding of organizations other than banks and financial intermediaries on the NCRP portal. It collects regulatory and organizational details required to verify and authorize access for such entities.

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### Registration for Other Entities

Please fill all required details to get onboarded

Select Other Category:

Other

**Regulatory Details**

Other Organization Name: \*

-- Type Organization Name --

Date of Incorporation

01 / 01 / 2026

RBI Registration Certificate

Browse... No file selected.

GST Certificate

Browse... No file selected.

FIU Registration ID

FIU Registration Screenshot

Browse... No file selected.

**Financial Information**

No balance sheet required (Entity age < 1 year).

Nodal Officer Sheet

Browse... No file selected.

Upload Terms & Conditions

Browse... No file selected.

Submit Registration

Reset





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## 8. NCRP Onboarding Dashboard Page for registered Users.

This page provides a consolidated dashboard for registered users to view onboarding status and submission details.

It also includes support features such as registration tracking and the NCRP help bot for user assistance.

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NCRP Dashboard

User ID: btech10726.24@bitmesra.ac.in Logout

Check Registration Status

Registration Summary

Category : BANK  
Submitted On : Thu Jan 22 2026  
Submission Status : Submitted  
Onboarding Status : Rejected

User Details

Full Name : Prashant Shah Gond  
Official Email : btech10726.24@bitmesra.ac.in  
User Email : prashant.gond100@gmail.com  
Designation : student  
Contact Number : 8001354567  
User ID : btech10726.24@bitmesra.ac.in  
Bank Category : SBI  
Bank Name : State Bank of India  
Incorporation Date : 2026-01-01  
FIU Registration ID : test123

Documents Uploaded

✓ RBI Certificate  
✓ GST Certificate  
✓ FIU Screenshot  
✓ Balance Sheet Year 1  
✓ Balance Sheet Year 2  
✓ Balance Sheet Year 3  
✓ Nodal Officer Sheet  
✓ Terms & Conditions

Admin Remarks

NCRP Help Bot

Hi 🤖 I'm the NCRP Help Bot.  
Select a question below to get instant help.

Which documents are required?

Required documents include:

- RBI Certificate
- GST Certificate
- FIU Screenshot
- Balance Sheets
- Nodal Officer Details
- Terms & Conditions

Which documents are required?

What if a document is missing?

How to contact support?

Onboarding login credentials



Admin Remarks

#1 nodal officer sheet upload

#2 re upload nodal sheet

[Edit & Resubmit](#)



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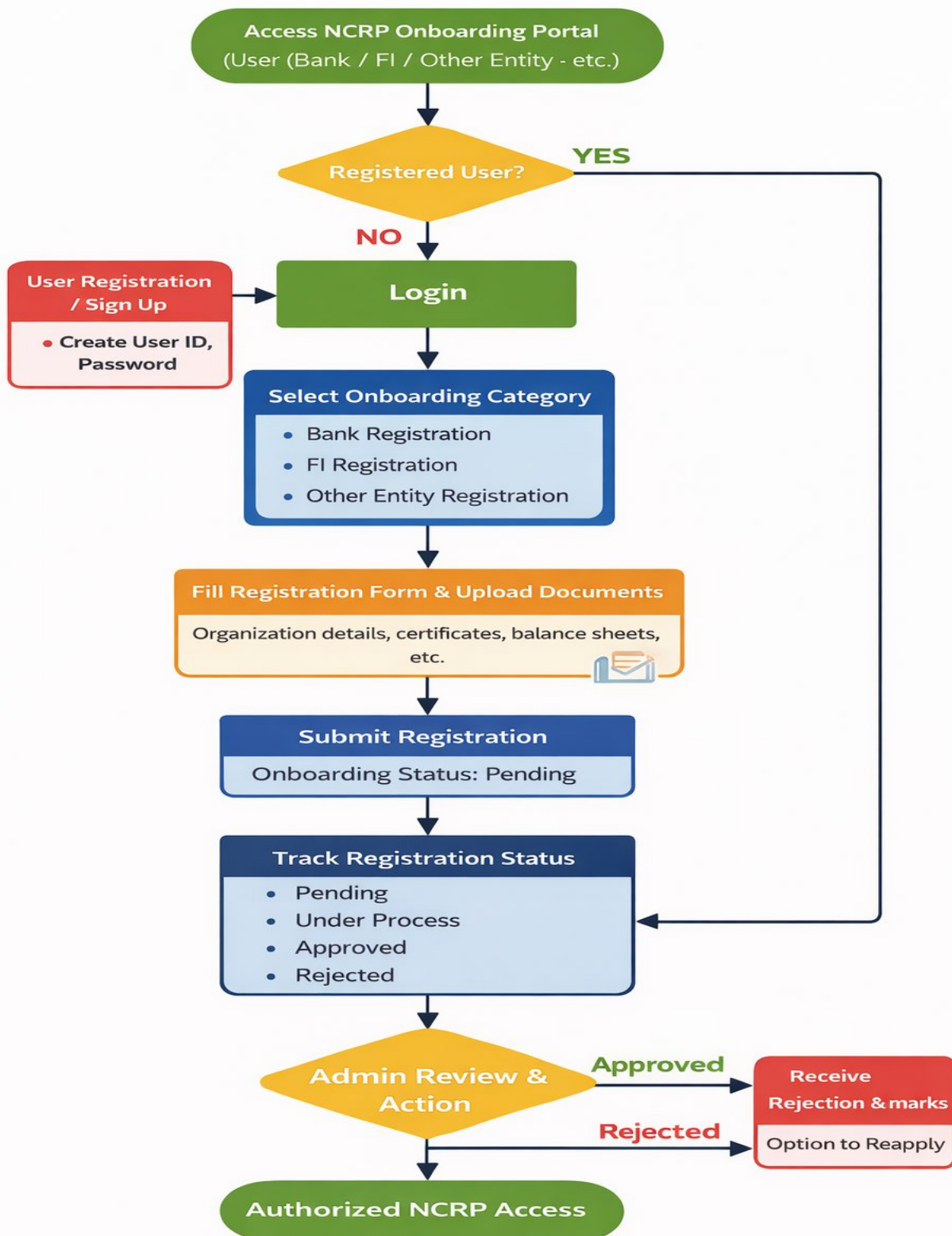


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# USER-SITE ONBOARDING WORKFLOW

## User-Side Onboarding Workflow - NCRP Portal



## **Chapter 3**

### **Implementation of Role-Based Access Control, Authentication Mechanisms, Document Verification, and Administrative Approval Workflows for Secure Financial Cyber Fraud Reporting and Management**

#### **3.1 Introduction**

The increasing scale and complexity of financial cyber fraud incidents demand a secure and well-governed access management system within the National Cyber Crime Reporting Portal (NCRP). As multiple stakeholders such as banks, financial intermediaries, other entities, and administrators interact with sensitive data, it becomes essential to implement strong access control, authentication, and verification mechanisms. This chapter discusses the implementation of Role-Based Access Control (RBAC), authentication mechanisms, document verification processes, and administrative approval workflows to ensure secure and efficient financial cyber fraud reporting and management.

#### **3.2 Need for Access Control and Secure Workflows**

Financial cyber fraud cases involve confidential information including personal identities, banking details, transaction records, and regulatory documents. Unauthorized access or misuse of such information can lead to serious legal and financial consequences. Therefore, NCRP requires a structured security framework that ensures:

- Access is granted strictly on a need-to-know basis
- Only verified users can perform specific actions
- Every activity is traceable and auditable
- Administrative oversight is maintained at all stages

RBAC and secure workflows together form the foundation of this framework.



### 3.3 Role-Based Access Control (RBAC)

Role-Based Access Control is a security mechanism where system access permissions are assigned based on the roles of users rather than individual identities. In the NCRP system, RBAC ensures that users can only access information and functionalities relevant to their designated role.

#### 3.3.1 Roles Defined in NCRP

The following key roles are implemented within the system:

- **Bank User:** Access limited to bank-related fraud cases, document uploads, and response actions
- **Financial Intermediary (FI) User:** Access to payment system related fraud information
- **Other Entity User:** Restricted access based on organizational function
- **Administrator:** Full access to onboarding requests, document verification, and approval workflows

Each role is mapped with predefined permissions to prevent unauthorized actions.

### 3.4 Authentication Mechanisms

Authentication ensures that only legitimate users gain access to the NCRP portal. The system employs multiple authentication layers to strengthen security.

#### 3.4.1 User Credential Authentication

Users are required to log in using a unique User ID and password created during the onboarding process. Password policies enforce minimum complexity requirements and periodic updates.

#### 3.4.2 One-Time Password (OTP) Authentication

To add an additional security layer, OTP-based verification is used during login and sensitive operations. OTPs are sent to registered contact details, reducing the risk of unauthorized access.



### **3.4.3 Secure Communication**

All authentication transactions are performed over encrypted channels to protect credentials from interception.

### **3.5 Document Verification Process**

Document verification plays a critical role in validating the authenticity of banks, financial intermediaries, and other entities seeking access to NCRP.

#### **3.5.1 Types of Documents Verified**

Entities are required to submit the following documents during onboarding:

- RBI registration certificate
- GST certificate
- FIU registration details
- Balance sheets and financial statements
- Nodal officer authorization documents
- Signed terms and conditions

#### **3.5.2 Verification Workflow**

Submitted documents are reviewed through the administrative interface. Administrators can view, verify, approve, or reject individual documents. This step ensures regulatory compliance and institutional legitimacy.

### **3.6 Administrative Approval Workflow**

The administrative approval workflow ensures that onboarding requests undergo proper scrutiny before access is granted.

#### **3.6.1 Request Status Categories**

Onboarding requests are categorized into the following statuses:

- Pending
- Under Process
- Approved
- Rejected

These categories allow administrators to track and manage requests efficiently.

### **3.6.2 Approval and Rejection Actions**

Administrators can approve or reject applications based on document verification results. In case of rejection, remarks are provided to inform users about deficiencies or corrections required.

### **3.7 Dashboard and Monitoring Features**

The admin dashboard provides a centralized view of all onboarding activities. It displays real-time statistics of pending, approved, and rejected requests, enabling effective monitoring and decision-making. Audit logs capture user activities such as login attempts, document access, approvals, and rejections, ensuring accountability and traceability.

### **3.8 Security Measures and Compliance**

The system follows best practices in cybersecurity and access governance. Key security measures include:

- Role-based access enforcement
- Encrypted data transmission
- Secure storage of credentials and documents
- Regular access reviews by administrators

These measures help NCRP comply with national cybersecurity standards and data protection policies.

### **3.9 Benefits of the Implemented System**

The implementation of RBAC and secure workflows provides multiple benefits:

- Reduced risk of unauthorized access
- Faster and more reliable onboarding of entities
- Improved coordination between stakeholders
- Enhanced transparency and accountability

The system strengthens the overall effectiveness of financial cyber fraud reporting and management.

### 3.10 Conclusion

The implementation of Role-Based Access Control, authentication mechanisms, document verification, and administrative approval workflows significantly enhances the security and governance of the NCRP portal. By ensuring controlled access and structured verification, the system supports secure collaboration among banks, financial intermediaries, other entities, and law enforcement agencies, thereby improving the national response to financial cyber fraud incidents.



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## 1. Admin Login Page

This page provides secure login access to NCRP administrators.

Only authorized admin users can log in to manage onboarding and verification processes.

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### Admin Login

Login

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## 2. Admin Dashboard Page

This dashboard displays a summary of onboarding requests categorized as pending, under process, approved, and rejected.

It enables administrators to monitor onboarding activities efficiently.

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Welcome, Admin

Logout



### Admin Dashboard

PENDING  
1

UNDER PROCESS  
2

APPROVED  
3

REJECTED  
4

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### 3. Registration Requests – Pending Page

This page lists onboarding requests that are pending administrative review.  
Admins can view submitted entity details and download uploaded documents.

#### Registration Requests

[Back to Dashboard](#)

| Bank Name                                                                                                       | Category | Status  | Action                                                    |
|-----------------------------------------------------------------------------------------------------------------|----------|---------|-----------------------------------------------------------|
| Mastercard Transaction Services (Canada) Inc.(formerly Transfast Inc., Canada and Global Foreign Exchange Inc.) | fi       | Pending | <a href="#">View Details</a> <a href="#">Download All</a> |

### 4. Registration Requests – Under Process Page

This page shows onboarding applications currently under verification.  
It helps administrators track requests that are being reviewed or validated.

#### Registration Requests

[Back to Dashboard](#)

| Bank Name                            | Category | Status        | Action                                                    |
|--------------------------------------|----------|---------------|-----------------------------------------------------------|
| American Express Banking Corporation | bank     | Under Process | <a href="#">View Details</a> <a href="#">Download All</a> |
| ESAF Small Finance Bank Limited      | bank     | Under Process | <a href="#">View Details</a> <a href="#">Download All</a> |

## 5. Registration Requests – Approved Page

This page displays entities whose onboarding requests have been approved. Approved entities are granted authorized access to the NCRP portal.

### Registration Requests

[Back to Dashboard](#)

| Bank Name                               | Category | Status   | Action                       |                              |
|-----------------------------------------|----------|----------|------------------------------|------------------------------|
| -                                       | bank     | Approved | <a href="#">View Details</a> | <a href="#">Download All</a> |
| Airtel Payments Bank Limited            | bank     | Approved | <a href="#">View Details</a> | <a href="#">Download All</a> |
| Saraswat Co-operative Bank Ltd., Bombay | bank     | Approved | <a href="#">View Details</a> | <a href="#">Download All</a> |

## 6. Registration Requests – Rejected Page

This page lists onboarding requests that have been rejected by the administrator. Rejected entries include entities that failed verification or compliance checks.

### Registration Requests

[Back to Dashboard](#)

| Bank Name                         | Category | Status   | Action                       |                              |
|-----------------------------------|----------|----------|------------------------------|------------------------------|
| -                                 | bank     | Rejected | <a href="#">View Details</a> | <a href="#">Download All</a> |
| CYBER FRAUD MITIGATION CENTRE     | other    | Rejected | <a href="#">View Details</a> | <a href="#">Download All</a> |
| Hitachi Payment Services Pvt. Ltd | fi       | Rejected | <a href="#">View Details</a> | <a href="#">Download All</a> |
| State Bank of India               | bank     | Rejected | <a href="#">View Details</a> | <a href="#">Download All</a> |



## 7. Registration Details and Document Verification Page

This page allows administrators to review complete registration details and uploaded documents. Admins can approve or reject individual documents and provide remarks to the user.



|                           |                                               |
|---------------------------|-----------------------------------------------|
| 1. CATEGORY               | bank                                          |
| 2. SUB-CATEGORY           | CO OP BANK-<br>SCHEDULED URBAN CO<br>OP BANKS |
| 3. BANK / FI / OTHER NAME | Saraswat Co-operative<br>Bank Ltd., Bombay    |
| 4. Official Full Name     | sneha singh                                   |
| 5. Official Email         | snehacmc@gmail.com                            |
| 6. User Email             | snehasingh@gmail.com                          |
| 7. Designation            | TA                                            |
| 8. Official Cont. No.     | 9999999999                                    |
| 9. Date of Incorporation  | 2026-01-01                                    |
| 10. FIU Registration Id   | ABCD1243                                      |

### Uploaded files

#### RBI\_CERTIFICATE

[VIEW DETAILS](#)

[APPROVE](#)

[REJECT](#)

#### GST\_CERTIFICATE

[VIEW DETAILS](#)

[APPROVE](#)

[REJECT](#)

#### FIU\_SCREENSHOT

[VIEW DETAILS](#)

[APPROVE](#)

[REJECT](#)

#### NODAL\_OFFICER\_SHEET

[VIEW DETAILS](#)

[APPROVE](#)

[REJECT](#)

#### TERMS\_CONDITIONS

[VIEW DETAILS](#)

[APPROVE](#)

[REJECT](#)

[APPROVE](#)

[REJECT](#)

### Remark to user

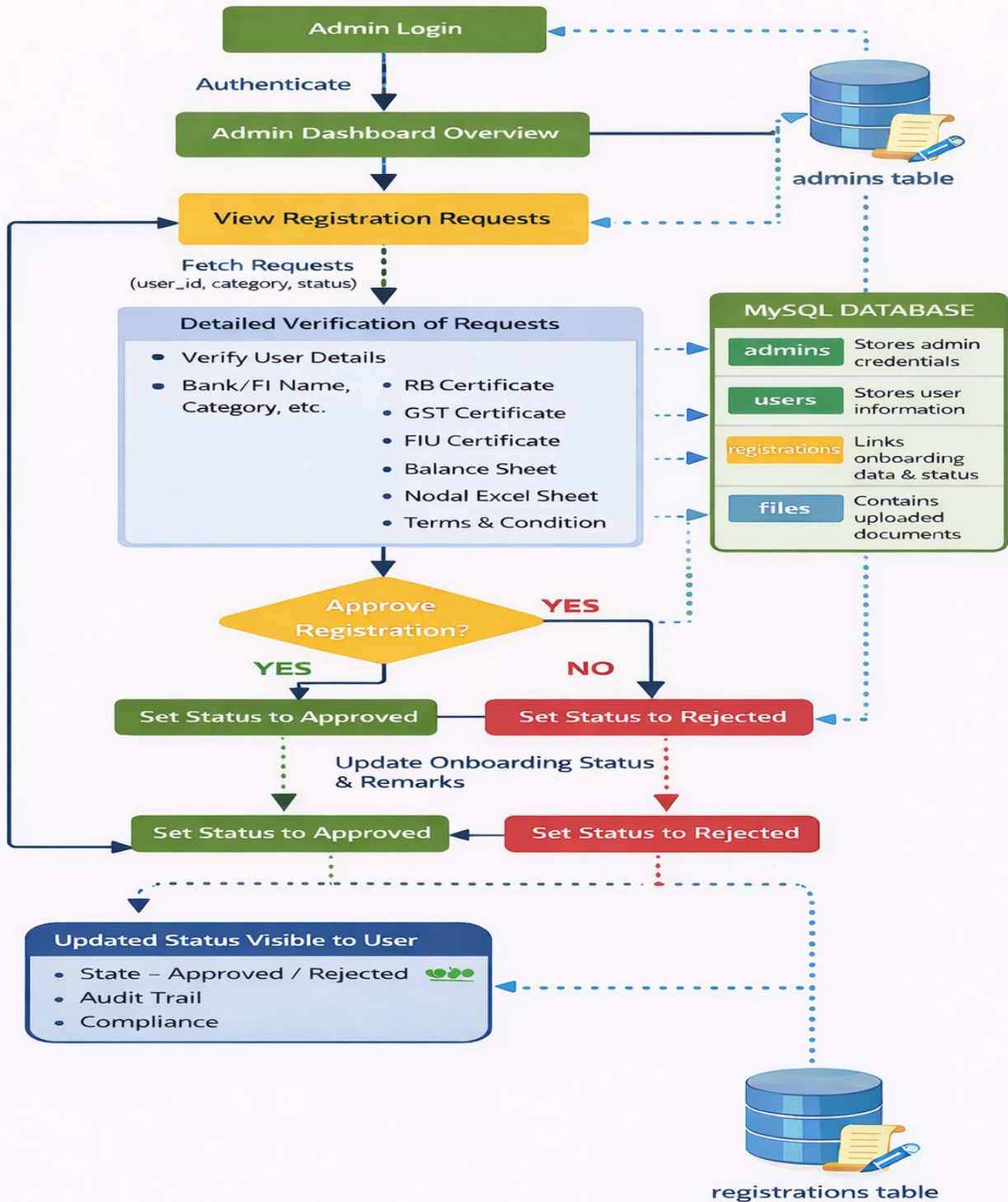
Type your remark for the user...

[SEND REMARK](#)

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# ADMIN-SITE ONBOARDING WORKFLOW

## Admin-Side Onboarding Workflow – NCRP Platform with MySQL Integration



## **Chapter 4**

### **Security Framework and Access Governance for Financial Cyber Fraud Management (MySQL Database Implementation)**

#### **4.1 Introduction**

Financial cyber fraud has emerged as one of the most critical challenges in the digital ecosystem, affecting citizens, banks, financial intermediaries, and government institutions. The National Cyber Crime Reporting Portal (NCRP) handles large volumes of sensitive data related to fraud complaints, user identities, financial documents, and investigation workflows. To safeguard this information, a robust security framework and access governance mechanism is essential.

This chapter explains the security framework and access governance model implemented for financial cyber fraud management using a MySQL database. The design focuses on secure data storage, controlled access, role-based authorization, and administrative oversight to ensure data confidentiality, integrity, and availability.

#### **4.2 Role of Database Security in NCRP**

The database forms the backbone of the NCRP system, storing critical information such as user credentials, onboarding requests, uploaded regulatory documents, and approval statuses. Any compromise at the database level can severely impact system security and trust.

MySQL is used as the relational database management system due to its reliability, support for strong constraints, and ability to enforce data integrity through relationships and access controls. Database-level security complements application-level security to provide end-to-end protection.

### **4.3 Database Schema Overview**

The MySQL database schema is designed to support secure onboarding and access governance. It consists of the following core tables:

- Users table
- Registrations table
- Files table
- Admins table

These tables are interconnected using primary and foreign keys to maintain referential integrity and prevent unauthorized or inconsistent data manipulation.

### **4.4 User Authentication and Credential Security**

#### **4.4.1 Users Table Design**

The users table stores registered user credentials including unique user IDs, encrypted passwords, and account creation timestamps. Each user is uniquely identified using an auto-incremented primary key, ensuring accurate mapping across the system.

Password fields are designed to store hashed values rather than plain text passwords. This protects user credentials even if database access is compromised.

#### **4.4.2 Secure Authentication Workflow**

During login, user credentials are validated against stored records in the MySQL database. Authentication success depends on correct credentials and role validation, ensuring only authorized users gain access.

### **4.5 Role-Based Access Control (RBAC) Using Database Design**

RBAC is implemented through a combination of database structure and application logic. Each user is assigned a role based on their category such as bank, financial intermediary, other entity, or administrator.

The registrations table includes a category field that defines the role and scope of access. This ensures that users can only access records and



functionalities relevant to their assigned role, enforcing the principle of least privilege.

#### **4.6 Secure Onboarding and Registration Management**

The registrations table stores onboarding data submitted by banks, FIs, and other entities. Registration data is stored in structured JSON format, allowing flexibility while maintaining consistency.

Each registration is linked to a specific user using foreign key constraints. This linkage ensures accountability and prevents orphan records or unauthorized registrations.

#### **4.7 Document Storage and Verification Security**

The files table stores regulatory and supporting documents uploaded during onboarding. Documents such as RBI certificates, GST certificates, balance sheets, and authorization letters are stored as BLOBs.

Each file record includes a file type and verification status, enabling document-level approval or rejection. Foreign key constraints ensure that documents are always linked to a valid registration.

#### **4.8 Administrative Access Governance**

Administrative users are managed through the admins table, which stores admin usernames and encrypted passwords. Only records present in this table are allowed to access the admin dashboard.

Admins have exclusive rights to view onboarding requests, verify documents, approve or reject registrations, and provide remarks. This separation of privileges prevents misuse of administrative functions.

#### **4.9 Workflow Status Tracking and Monitoring**

The system uses status fields such as pending, under process, approved, and rejected to track onboarding progress. These statuses are stored directly in the MySQL database, ensuring consistency across the system.

Status tracking enables real-time monitoring of onboarding requests and supports transparent decision-making by administrators.

#### **4.10 Data Integrity and Referential Constraints**

MySQL constraints such as primary keys, foreign keys, and cascading deletes are used to enforce data integrity. For example, deleting a registration automatically removes associated documents, preventing stale data.

These constraints protect the system from inconsistent states and accidental data corruption.

#### **4.11 Auditability and Accountability**

Timestamps such as `created_at` and `uploaded_at` are stored for users, registrations, and files. These fields help track when records were created or modified.

Auditability supports forensic analysis during cybercrime investigations and enhances accountability among system users.

#### **4.12 Compliance with Security Best Practices**

The database design aligns with security best practices such as least privilege access, encrypted credential storage, and strict role separation. Sensitive data is accessed only by authorized roles, reducing exposure risk.

Such compliance strengthens trust among stakeholders and supports lawful handling of financial cyber fraud data.

#### **4.13 Benefits of the MySQL-Based Security Framework**

Using MySQL for implementing security and access governance offers several advantages:

- Strong relational integrity
- Efficient role-based access enforcement
- Secure document management
- Scalable onboarding workflow management

These benefits improve the overall robustness of the NCRP system.



#### **4.14 Limitations and Future Enhancements**

Although the current MySQL-based framework provides strong security, future enhancements can include database encryption at rest, advanced audit logging, and integration with centralized identity management systems. Such improvements can further strengthen the NCRP security posture.

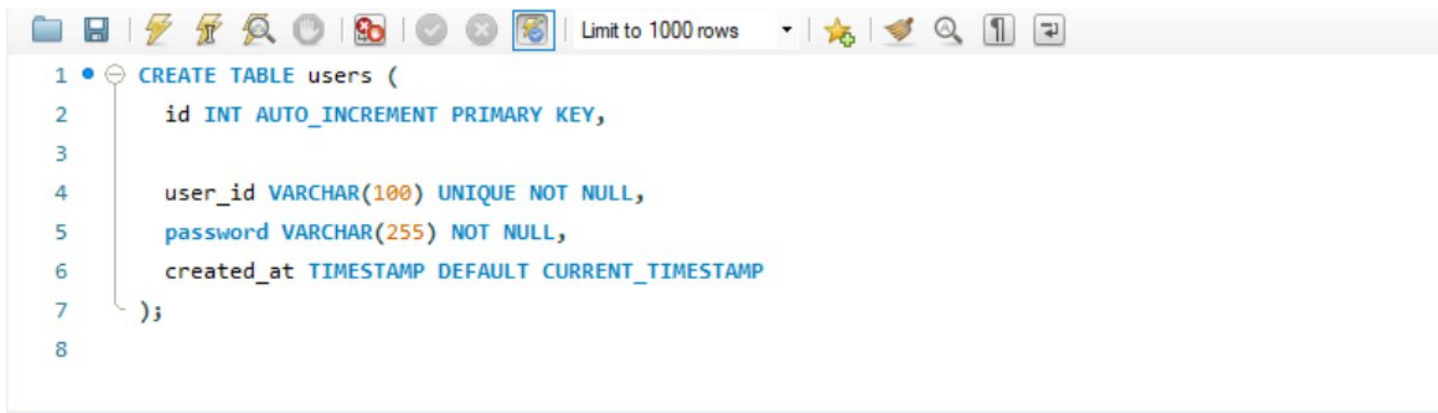
#### **4.15 Conclusion**

The implementation of a security framework and access governance model using a MySQL database plays a crucial role in secure financial cyber fraud management within the NCRP portal. By combining structured database design, role-based access control, secure authentication, document verification, and administrative oversight, the system ensures confidentiality, integrity, and accountability. This approach significantly enhances the effectiveness of financial cyber fraud reporting and management at a national level.

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## 1. Users Table Creation

This table stores login credentials of registered users accessing the NCRP onboarding system. Each user is assigned a unique User ID with password details and account creation timestamp. It supports secure authentication and identity management.



```
1 • CREATE TABLE users (  
2     id INT AUTO_INCREMENT PRIMARY KEY,  
3  
4     user_id VARCHAR(100) UNIQUE NOT NULL,  
5     password VARCHAR(255) NOT NULL,  
6     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
7 );  
8
```

## 2. Registrations Table Creation

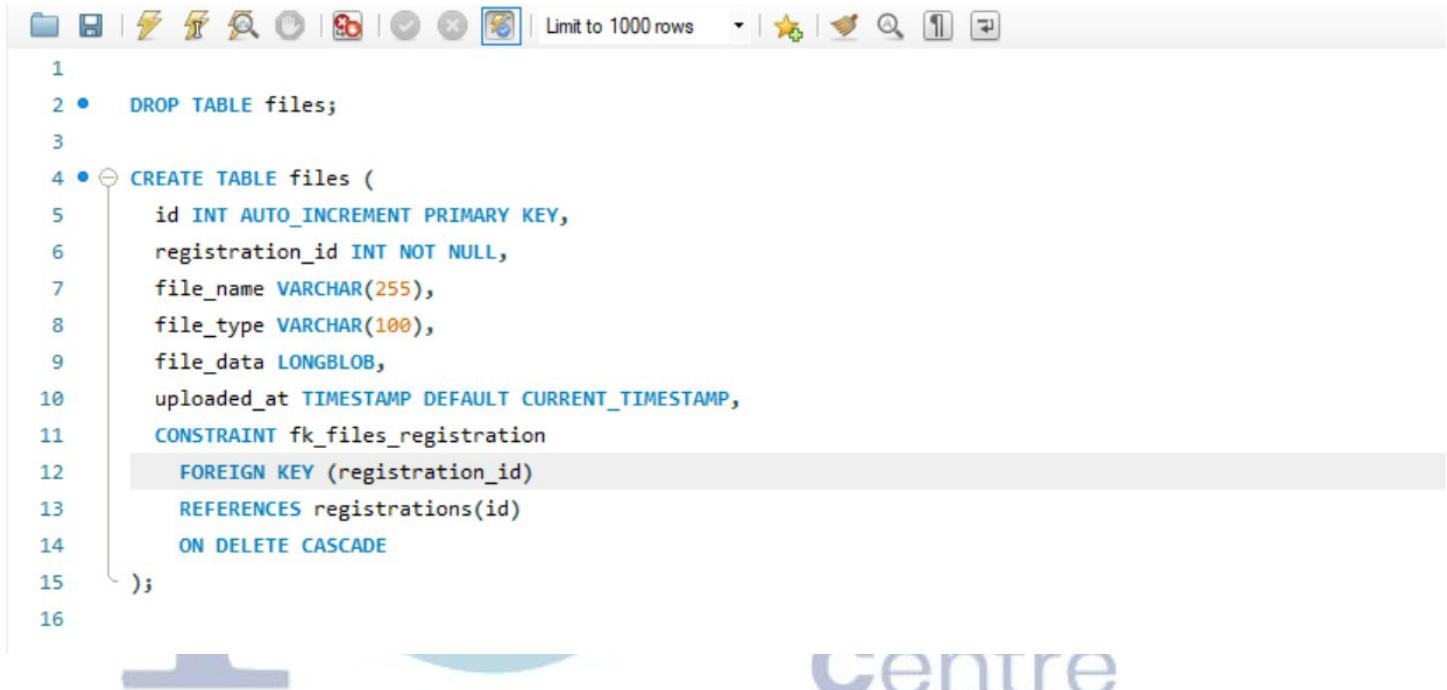
This table stores onboarding requests submitted by banks, financial intermediaries, and other entities. It links each registration to a user and stores category-specific form data in JSON format. The table enables tracking of onboarding requests through timestamps and status fields.



```
1 • CREATE TABLE registrations (  
2     id INT AUTO_INCREMENT PRIMARY KEY,  
3     user_id INT NULL,  
4     category ENUM('bank', 'fi', 'other') NOT NULL,  
5     form_data JSON,  
6     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
7     FOREIGN KEY (user_id) REFERENCES users(id)  
8 );  
9
```

### 3. Files Table Creation

This table stores all regulatory and supporting documents uploaded during onboarding. Each file is linked to a specific registration and includes file type and upload timestamp. Cascade deletion ensures data integrity when a registration is removed.



```
1
2 • DROP TABLE files;
3
4 • CREATE TABLE files (
5     id INT AUTO_INCREMENT PRIMARY KEY,
6     registration_id INT NOT NULL,
7     file_name VARCHAR(255),
8     file_type VARCHAR(100),
9     file_data LONGBLOB,
10    uploaded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
11    CONSTRAINT fk_files_registration
12    FOREIGN KEY (registration_id)
13    REFERENCES registrations(id)
14    ON DELETE CASCADE
15 );
16
```

### 4. Admins Table Creation

This table maintains credentials for NCRP administrative users. Only authorized administrators stored in this table can access the admin dashboard. It supports secure login and administrative control functions.



```
1 • CREATE TABLE admins (
2     id INT AUTO_INCREMENT PRIMARY KEY,
3     username VARCHAR(100) UNIQUE NOT NULL,
4     password VARCHAR(255) NOT NULL,
5     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
6 );
7
```

## 5. Users Table – Stored Records

This displays registered users with their User IDs and creation timestamps. It shows both entity users and administrative users stored in the system. The data confirms successful user registration and account creation.

| Result Grid | Filter Rows:                 | Edit:      | Export/Import:      | Wrap Cell Content: |
|-------------|------------------------------|------------|---------------------|--------------------|
| id          | user_id                      | password   | created_at          |                    |
| 1           | btech10726.24@bitmesra.ac.in | 8001354567 | 2026-01-22 00:14:35 |                    |
| 2           | lavishcfmc@gmail.com         | 8001354567 | 2026-01-22 00:18:13 |                    |
| 3           | priyankacfmc@gmail.com       | 8001354567 | 2026-01-22 00:22:17 |                    |
| 4           | snehacfmc@gmail.com          | 8001354567 | 2026-01-22 00:25:26 |                    |
| 5           | prashant.gond100@gmail.com   | 8001354567 | 2026-01-22 02:48:03 |                    |
| 6           | lavishyay@gmail.com          | 123456     | 2026-01-23 14:49:06 |                    |
| 7           | akhileshcfmc@gmail.com       | 8001354567 | 2026-01-23 17:57:05 |                    |
| 8           | helloworld                   | 8001354567 | 2026-01-23 23:56:22 |                    |
| 9           | admin@ncrp                   | admin123   | 2026-01-24 18:53:21 |                    |
| 10          | director@cfmc                | 8001354567 | 2026-01-24 19:07:20 |                    |
| NULL        | NULL                         | NULL       | NULL                |                    |

users 7 × registrations 8 files 9

## 6. Registrations Table – Stored Records

This table shows onboarding requests submitted under different categories such as bank, FI, and other. Each record reflects the current onboarding status like approved, rejected, pending, or under process. It supports workflow tracking and administrative decision-making.

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

|   | id   | user_id | category | form_data                                              | created_at          | onboarding_status |
|---|------|---------|----------|--------------------------------------------------------|---------------------|-------------------|
| ▶ | 1    | 1       | bank     | {"full_name": "Prashant Shah Gond", "user_ema...       | 2026-01-22 00:14:35 | Rejected          |
|   | 2    | 2       | fi       | {"fi_name": "Hitachi Payment Services Pvt. Ltd"...     | 2026-01-22 00:18:13 | Rejected          |
|   | 3    | 3       | other    | {"full_name": "priyanka kumari", "user_email": "...    | 2026-01-22 00:22:17 | Rejected          |
|   | 4    | 4       | bank     | {"full_name": "sneha singh", "user_email": "sne...     | 2026-01-22 00:25:26 | Under Process     |
|   | 5    | 5       | bank     | {"full_name": "zippy Luck", "user_email": "prasha...   | 2026-01-22 02:48:03 | Approved          |
|   | 6    | 6       | bank     | {"full_name": "abcsde", "user_email": "abc@gm...       | 2026-01-23 14:49:06 | Rejected          |
|   | 7    | 7       | bank     | {"full_name": "akhilesh sir", "user_email": "akhile... | 2026-01-23 17:57:05 | Under Process     |
|   | 8    | 8       | bank     | {"full_name": "helloworld", "user_email": "hellow...   | 2026-01-23 23:56:22 | Approved          |
|   | 9    | 9       | bank     | {"full_name": "ppp", "user_email": "admin@gmai...      | 2026-01-24 18:53:21 | Under Process     |
|   | 10   | 10      | fi       | {"fi_name": "Mastercard Transaction Services (...      | 2026-01-24 19:07:20 | Pending           |
| ✱ | NULL | NULL    | NULL     | NULL                                                   | NULL                | NULL              |



## 7. Files Table – Stored Records

This table displays uploaded documents such as RBI certificates, GST certificates, and balance sheets.

Each document includes its verification status indicating approval or pending review.

It supports document-level validation within the onboarding workflow.

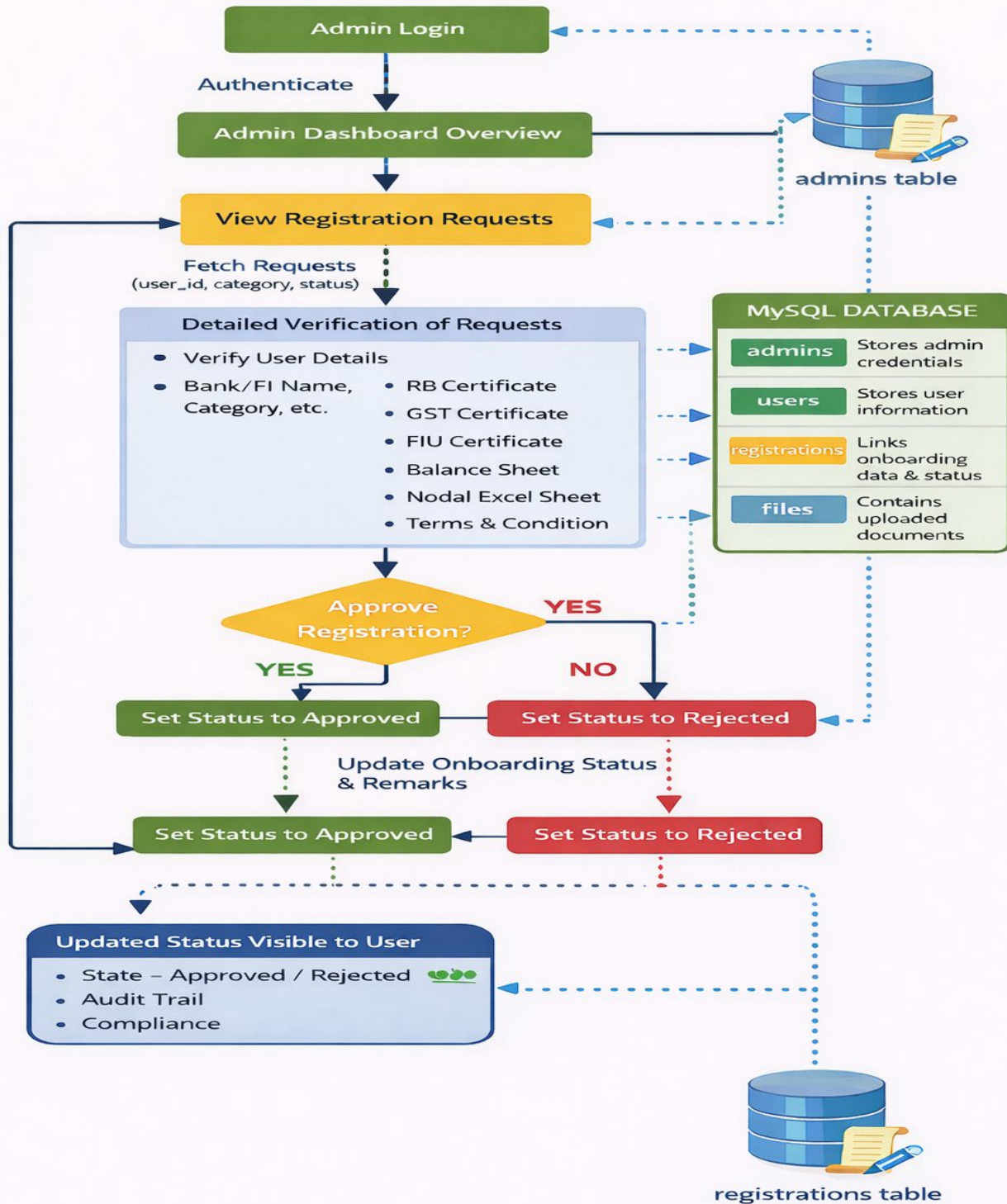
| Result Grid                       |    |                 |                                                |                      |                |                     |          |
|-----------------------------------|----|-----------------|------------------------------------------------|----------------------|----------------|---------------------|----------|
|                                   |    | Filter Rows:    | Edit:                                          |                      | Export/Import: | Wrap Cell Content:  |          |
|                                   | id | registration_id | file_name                                      | file_type            | file_data      | uploaded_at         | status   |
| ▶                                 | 1  | 1               | RBI licence copy.pdf                           | rbi_certificate      | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 2  | 1               | GST NEW REG NO..pdf                            | gst_certificate      | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 3  | 1               | FIU LOGIN SCREEN.jpg                           | fiu_screenshot       | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 4  | 1               | CDCCB BALSHT 31.03.2022.PDF                    | balance_sheet_year_1 | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 5  | 1               | CDCCB BALSHT 31.03.2023.pdf                    | balance_sheet_year_2 | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 6  | 1               | CDCCB BALSHT 31.03.2024.pdf                    | balance_sheet_year_3 | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 7  | 1               | Onboarding Format_Chitradurga DCC Bank Ltd.... | nodal_officer_sheet  | BLOB           | 2026-01-22 00:16:29 | Rejected |
|                                   | 8  | 1               | Terms & Conditions _ NCRP_CFCFRMS_ Chitrad...  | terms_conditions     | BLOB           | 2026-01-22 00:16:29 | Approved |
|                                   | 9  | 2               | RBI licence copy.pdf                           | rbi_certificate      | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 10 | 2               | GST NEW REG NO..pdf                            | gst_certificate      | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 11 | 2               | FIU LOGIN SCREEN.jpg                           | fiu_screenshot       | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 12 | 2               | CDCCB BALSHT 31.03.2023.pdf                    | balance_sheet_year_1 | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 13 | 2               | CDCCB BALSHT 31.03.2024.pdf                    | balance_sheet_year_2 | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 14 | 2               | Onboarding Format_Chitradurga DCC Bank Ltd.... | nodal_officer_sheet  | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 15 | 2               | Terms Conditions _ NCRP_CFCFRMS_ Chitradu...   | terms_conditions     | BLOB           | 2026-01-22 00:19:51 | Pending  |
|                                   | 16 | 3               | RBI licence copy.pdf                           | rbi_certificate      | BLOB           | 2026-01-22 00:23:29 | Approved |
|                                   | 17 | 3               | GST NEW REG NO..pdf                            | gst_certificate      | BLOB           | 2026-01-22 00:23:29 | Approved |
|                                   | 18 | 3               | FIU LOGIN SCREEN.jpg                           | fiu_screenshot       | BLOB           | 2026-01-22 00:23:29 | Approved |
| users 7 registrations 8 files 9 x |    |                 |                                                |                      |                |                     |          |

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# MYSQL-DATABASE WORKFLOW

## Admin-Side Onboarding Workflow – NCRP Platform with MySQL Integration



# Summary

The present project focuses on the design, development, and implementation of a secure and integrated framework to support financial cyber fraud reporting and management through the National Cyber Crime Reporting Portal (NCRP), an initiative of the Government of India under the Ministry of Home Affairs. With the rapid growth of digital payments and online financial services, cyber fraud incidents such as phishing, UPI frauds, debit and credit card frauds, identity theft, and unauthorized transactions have increased significantly. NCRP serves as a centralized digital platform that enables citizens to report cybercrimes efficiently while assisting law enforcement agencies in investigation and response.

The primary objective of this project is to strengthen the institutional side of the NCRP ecosystem by developing a secure User ID creation and onboarding system for banks, financial intermediaries, and other authorized entities. The system ensures that only verified and legitimate organizations are granted access to sensitive fraud-related information. By incorporating structured onboarding workflows, document verification, and administrative approval mechanisms, the project enhances trust, accountability, and coordination among all stakeholders involved in financial cyber fraud management.

The initial phase of the project involved understanding the operational structure and functional workflow of the NCRP portal. This included studying how cybercrime complaints are registered by users, categorized based on the nature of the offence, and routed to appropriate state or central cybercrime units. The role of banks and financial intermediaries in validating transactions, freezing fraudulent accounts, and supporting investigations was also analyzed. This understanding formed the foundation for designing a secure onboarding and access governance framework aligned with NCRP objectives.

A major component of the project is the implementation of Role-Based Access Control (RBAC) to ensure controlled and restricted access to the system. Different roles such as bank users, financial intermediary users, other entity users, and administrators were defined, each with specific permissions. RBAC enforces the principle of least privilege, ensuring that users can access only the data and functionalities relevant to their

responsibilities. This significantly reduces the risk of unauthorized access and misuse of sensitive information.

Authentication and identity management mechanisms were incorporated to further strengthen system security. Each user is assigned a unique User ID during the onboarding process, and secure login procedures are enforced. Additional security measures such as OTP-based verification and credential validation help protect the system from unauthorized access attempts. These mechanisms collectively enhance the confidentiality and integrity of the NCRP portal.

The security framework and access governance model were implemented using a MySQL database, which acts as the core data repository for the system. Well-structured database tables such as users, registrations, files, and admins were designed to store user credentials, onboarding details, uploaded regulatory documents, and administrative information. Primary keys, foreign keys, and referential constraints were used extensively to maintain data integrity and prevent inconsistent or unauthorized data manipulation.

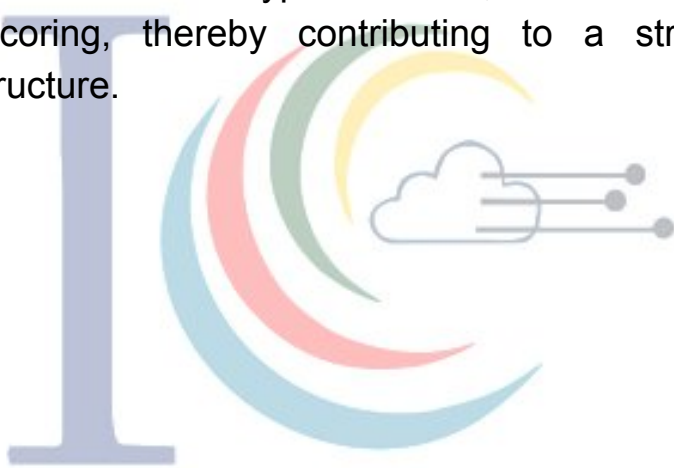
The registrations table plays a crucial role in managing onboarding requests submitted by banks, financial intermediaries, and other entities. Each registration record captures category-specific information and tracks the onboarding status using predefined states such as pending, under process, approved, and rejected. This status-based workflow ensures transparency and enables administrators to monitor and manage onboarding requests effectively.

Document management and verification were handled through a dedicated files table, where regulatory and supporting documents such as RBI certificates, GST certificates, balance sheets, FIU details, and authorization letters are securely stored. Each document is linked to its corresponding registration and includes verification status information. This allows administrators to approve or reject documents individually and maintain a clear audit trail for compliance and accountability.

Administrative users are managed separately through the admins table, ensuring a clear separation of privileges between normal users and administrators. Admin dashboards provide a consolidated view of onboarding activities, enabling efficient review, verification, approval, and rejection of requests. The inclusion of timestamps and status

indicators supports traceability and forensic analysis, which is essential in financial cyber fraud investigations.

Overall, this project demonstrates a practical and secure approach to financial cyber fraud management through effective access governance and database-driven security mechanisms. By integrating structured onboarding, role-based access control, secure authentication, document verification, and administrative oversight, the system strengthens the operational effectiveness of the NCRP portal. The proposed framework is scalable and can be further enhanced in the future by incorporating advanced security features such as encryption at rest, centralized identity management, and automated risk scoring, thereby contributing to a stronger national cybercrime response infrastructure.



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## **References**

### **1. Official Portals & Government Resources**

- National Cyber Crime Reporting Portal (NCRP), Government of India.  
<https://www.cybercrime.gov.in>
- Ministry of Home Affairs (MHA) – Indian Cyber Crime Coordination Centre (I4C).  
<https://www.mha.gov.in>
- Reserve Bank of India (RBI). Registered Banks/Fi's/o and Digital Payment Guidelines.  
<https://www.rbi.org.in>

### **2. Technical & Security Documentation**

- MDN Web Docs (Mozilla). HTML, CSS, and JavaScript Documentation.  
<https://developer.mozilla.org>
- OWASP Foundation. OWASP Top 10 – Web Application Security Risks.  
<https://owasp.org>
- MySQL Documentation. User Management and Database Security.  
<https://dev.mysql.com/doc>

### **3. Organizational Reference**

- Indian Cyber Crime Coordination Centre (I4C). Internal SOPs and onboarding workflow documentation.